

Surface stress and the narrow-ring β -plane analogue in WC25 turbulence

Velocity notation: $u'_{\text{rms}}(t) = \sqrt{\langle |u'_L(t)|^2 \rangle}$ denotes the total rms velocity fluctuation, not one component; $u'_{\text{rms},0}$ is its initial value. The stress ratio is $u^*/u'_{\text{rms},0}$. We reserve q for potential vorticity. Archived figure versions may retain q_0 or q' for these rms speeds; current figures use the updated notation.

Results document for the two October 2026 campaigns that extend Wagner & Constantinou (2025, JFM 1020 A51, "WC25"). Campaign A asks whether a constant along-wave surface stress turns the depth-alternating jets of initially Lagrangian-shear-free wave-averaged turbulence into Langmuir rolls. Campaign B asks whether the two-dimensional (x – z) WC25 dynamics started from the Vallis & Maltrud (1993) spectral ring reproduce the β -plane dumbbell spectrum and zonal organisation of Vallis (2017, Figs 12.3–12.4). Both campaigns ran on Delta AI GH200 GPUs with Oceananigans 0.107; every figure is Julia/CairoMakie output from the campaign analysis scripts. The two source reports are provided alongside this synthesis.

Summary

A • Zero stress shows WC25-like structures

A counter-wave surface jet ($-3.2e-3$) and depth-alternating interior bands grow from an initially shear-free Lagrangian field; their share of the relative energy rises from 0.19 ($t = 400$) to 0.40 ($t = 1000$) while the wave-free control stays below 0.02. The generated mean flow is weak relative to the imposed Stokes drift, so $U_E \approx -u^s$ remains a useful approximation; the Lagrangian mean does have jet shear.

A • Stress ratio organises the sampled cases

In the cases run, rolls appeared when the instantaneous ratio $u^*/u'_{\text{rms}}(t)$ reached ≈ 0.7 – 1 and only for $La \lesssim 0.3$: $u^*/u'_{\text{rms},0} = 0.01$ stayed indistinguishable from zero stress to $t = 1000$, 0.03 transitioned at $t \approx 277$, 0.1 at $t \approx 80$, and 0.3 – 1 gave shear turbulence with roll anisotropy ≤ 1.2 , the same as their wave-free controls. These are observations in this sampled setup (one seed at most ratios, two grids at three ratios), not necessary or sufficient conditions.

B • The dumbbell phenomenology is reproduced qualitatively

With shallow waves ($\beta_{\text{eff}} = 1/2$ exactly) the ring energy redistributes anisotropically and piles up near the k_z axis; the dashed global Rhines estimate does not quantitatively trace the depleted region: anisotropy index $0 \rightarrow 0.70$, centroid $9 \rightarrow 5.1$ cycles, x -elongated vorticity bands, a measured inverse energy flux. The six-seed $k_0 = 12$, $rb = 2$ ensemble shows the VM93 Fig. 5 dumbbell shape cleanly; this is a source-matched qualitative comparison, not an exact reproduction.

B • Weaker beta gives stronger mean jets in these runs

The exact-zonal energy fraction falls from 0.16 ($rb = 0.5$) to about 0.04 ($rb = 1$) and 0.017 ($rb = 2$) at 100 T_e ; ring width, a second grid (512^2), seed and the WENO regime give the same picture. Deep waves confine β dynamics to the upper third and add a seed-sensitive x -mean jet.

Cost

12.94 billed GPU-hours in total (surface stress 10.83, Vallis-2D 2.11) against a 60 GPU-hour ceiling; this includes validation, retries and analysis.

Methods

Shared numerics

WC25 configuration: wave-averaged (Craig–Leibovich) Navier–Stokes for the Lagrangian-mean velocity u^L in a unit domain, periodic in x (and y in 3-D), impermeable free-slip top ($z = 1$) and bottom, WENO(order 9) implicit LES with no explicit closure, RK3, Float64, Oceananigans 0.107. Stokes drift from the printed WC25 equations: deep $\partial z u^s = \frac{1}{4}e^{8(z-1)}$ ($u^s(1) = 1/32$), shallow/medium $\partial z u^s = z/2$. u^s is never added to the initial condition, so the primary initial Lagrangian mean is shear-free and the Eulerian mean starts at $-u^s$. The repository driver's DeepStokesShear(2, 8) profile is not eq. (2.7) and was not used by either campaign. All JLD2 output uses stream IO (memory-mapped IO fails on Lustre), checkpoints are segmented and were verified to restart bitwise identically.

	A · SURFACE STRESS (3-D)	B · VALLIS RING (2-D, X-Z)
Grid	128 ³ (sweep), 256 ³ (three cases); unit cube	256 ² (all), 512 ² (three cases); Flat y , $v \equiv 0$
Initial condition	Seeded solenoidal random field, spectrum $\propto (k/k_e)^4 e^{-2(k/k_e)^2}$, $k_e = 2\pi \cdot 10$, rms $u'_{rms,0} = 0.03$, zero horizontal means; WC25's 1000 $\rightarrow$ 10 vorticity spin-up skipped by design	Streamfunction ring $\psi = \sum 2\text{Re}[A e^{i2\pi m x}] \sin(n\pi z)$, Gaussian $E(K) \propto e^{-(K-K_0)^2/2\sigma^2}$, $K_0 = 2\pi \cdot 9$ (or 12), $\sigma =$ one mode spacing (a chosen width: VM93's $Z/E = 2K^2$ is consistent with a narrow ring but does not fix σ uniquely), exactly solenoidal on the C-grid, $k_x = 0$ modes removed (they would carry 6% of the energy)
Control parameter	$u^*/u'_{rms,0} \in \{0, 0.01, 0.03, 0.1, 0.3, 1\}$, $\tau = u^{*2}$ as a flux condition on u at $z = 1$; $La = \sqrt{(u^*/u^s(1))}$	$rb = \beta/(u'_{rms,0}K_0^2)$ with $\beta = \partial z z u^s = \frac{1}{2}$ (shallow); $rb = 1 \Rightarrow u'_{rms,0} = 1.56e-4$, turnover $T_e = 1/(u'_{rms,0}K_0) = 113 =$ Rossby time K_0/β
Families	deep waves (primary), medium waves ($\partial z u^s = z/2$), wave-free controls from the same Lagrangian field	shallow (β -plane exact), deep ($\beta(z) = 2e^{8(z-1)}$), surface-intensified), wave-free, plus a doubly periodic pseudo-spectral β -plane reference
Durations	$t = 400$ all; 1000 for deep $r \leq 0.3$, $r = 1$ and the $r = 0$ control	200 T_e pilots; 100 T_e sensitivity; 60 T_e ensemble
Validation	divergence, momentum ($1e-19$), $d\langle U \rangle/dt = \tau/H$, wave-only quiescence, energy monotone, budget residual ≤ 0 , restart	divergence $7e-18$, Parseval-exact spectra, linear Rossby wave within 0.23% with $-x$ phase, quiescence, Stokes work $6e-17$, Δt convergence to $1e-4$ for $\Delta t \leq 4$, restart

Physics of the 2-D analogue

With $\omega_y = \partial z u - \partial x w$, the curl of the wave-averaged momentum equation gives $D(\omega_y - \partial z u^s)/Dt = 0$, so $D\omega_y/Dt = (\partial z z u^s) w$: a β -plane with $z \leftrightarrow y$, $w \leftrightarrow v$, $\zeta = -\omega_y$ and $\beta_{\text{eff}}(z) = \partial z z u^s$. The Stokes momentum terms $\partial z u^s$ ($w \hat{x} - u \hat{z}$) are perpendicular to u^L and do no work on it; energy is conserved up to the implicit dissipation. For shallow waves $\beta_{\text{eff}} = \frac{1}{2}$ is uniform and the wall problem is the odd-symmetric subspace of a doubly periodic 1×2 β -plane; for deep waves β_{eff} varies by 3000 $\times$ over the depth and the case is not a homogeneous- β reproduction.

Case tables

A • Surface stress

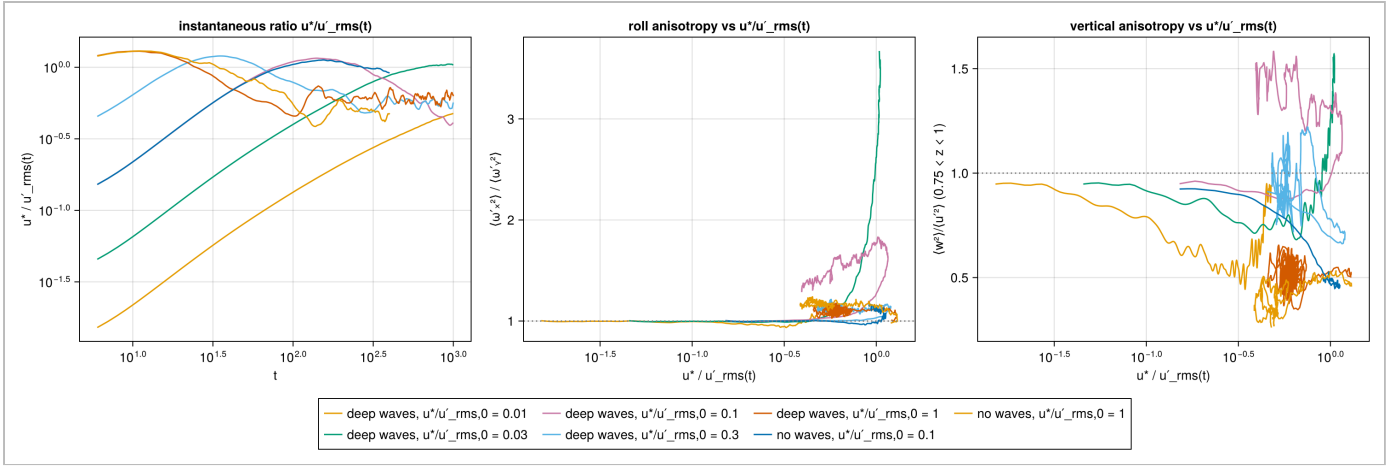
U*/U' _{RMS,0}	U*	LA	FAMILY	N	SEEDS	T_END	OUTCOME	
0	0	0	(deep, medium); undefined (none)	deep · none · medium	128 ³ , 256 ³ (deep)	1, 2	1000	WC25 regime: counter-wave surface jet, alternating bands, K_jet/K_rel 0.19 → 0.40
0.01	3e−4	0.10	deep	128 ³	1	1000	indistinguishable from zero stress (u*/u' _{rms} ≤ 0.3)	
0.03	9e−4	0.17	deep	128 ³ , 256 ³	1, 2	1000	transition at t ≈ 277 (u*/u' _{rms} = 0.74; 256 ³ : 0.66–0.74), Langmuir state by t = 1000	
0.1	3e−3	0.31	deep · none · medium	128 ³ , 256 ³ (deep)	1	400 (1000 deep)	rolls from t ≈ 80 (u*/u' _{rms} = 1.0; 256 ³ onset t = 156, ratio 1.06)	
0.3	9e−3	0.54	deep	128 ³	1	1000	shear-dominated, roll anisotropy ≤ 1.2	
1	3e−2	0.98	deep · none	128 ³	1, 2	400; deep extension to t ≈ 1000 still finishing at report time	shear turbulence, same anisotropy as its control; surface velocity O(1), where the implicit dissipation is velocity-dependent	

B • Vallis-2D

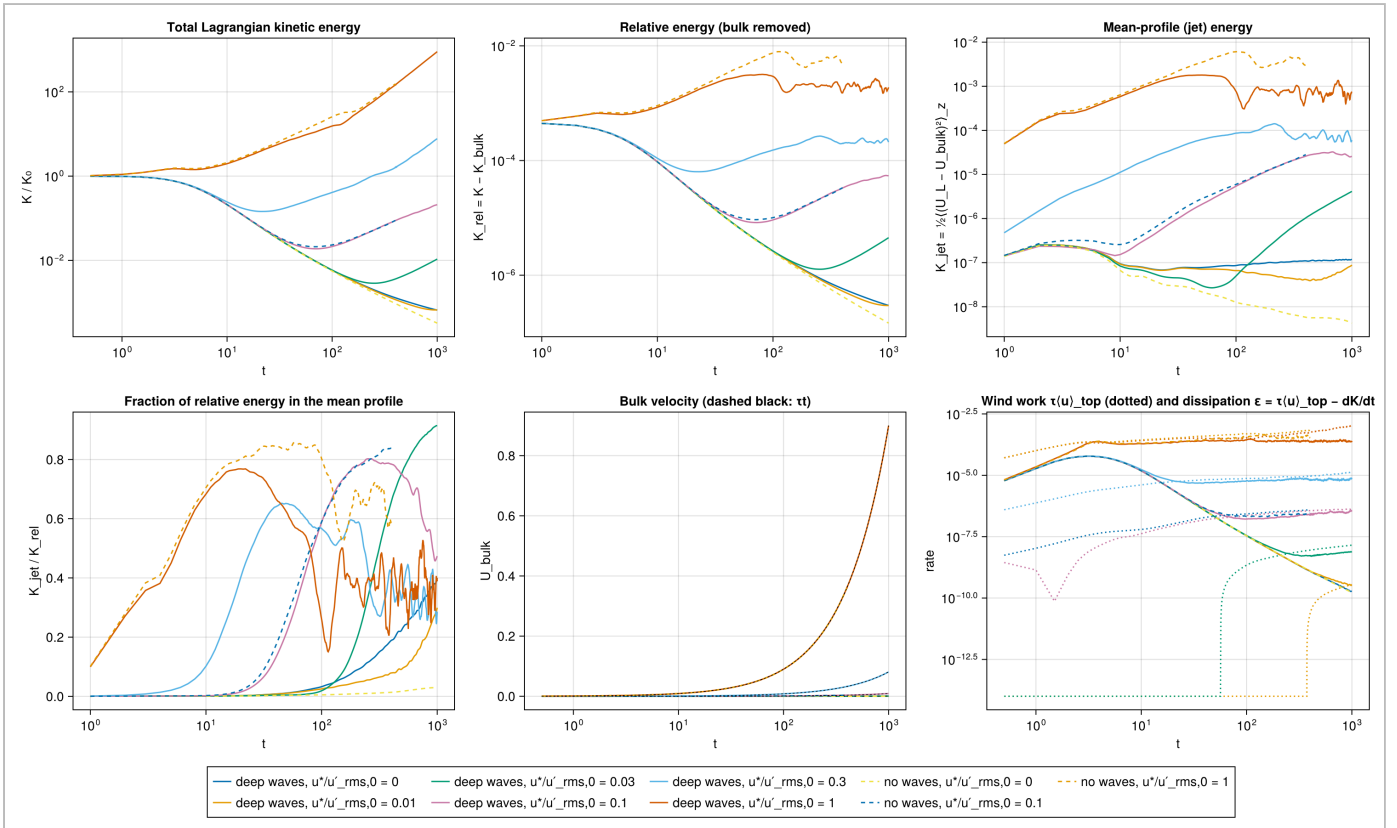
CASE SET	PARAMETERS	N	SEEDS	T_E	OUTCOME (EXACT-ZONAL K_JET/K · ANISOTROPY AT END)
pilots shallow / deep / none	$k_0 = 9$, $\sigma = 1$, $rb = 1$	256 ² , 512 ²	1	200	0.050 · 0.70 / 0.39 · 0.62 / 0.12 · 0.01 (512 ² : 0.050 · 0.715 / 0.384 / 0.133)
β -plane reference	doubly periodic, $\beta =$ $\frac{1}{2}$, same ring law and $u'_{rms,0}$, ∇^6 hyperviscosity	256 ²	1	200	0.060; same dumbbell as the wall case
rb bracket (shallow)	$rb = 0.5$, 2	256 ²	1	100	0.159 · 0.44 and 0.017 · 0.70 (base at 100 T_e $\approx$ 0.04 · 0.62)
ring width (shallow)	$\sigma = 0.5$, 2	256 ²	1	100	0.027 · 0.61 and 0.040 · 0.56
$k_x = 0$ modes kept	+6% initial zonal energy	256 ²	1	100	0.097 · 0.70
$\lambda = 200$ rescaling (shallow, none)	u^s and $u'_{rms,0} \times 200$, $u'_{rms,0} = 0.031$: identical physics, nonlinear WENO regime	256 ²	1	100	0.039 · 0.63 (K/K ₀ 0.990) and 0.118 · 0.01
second seed (shallow, none, deep)	$k_0 = 9$, $rb = 1$	256 ²	2	100	0.043 · 0.64 / 0.113 · 0.01 / 0.074 · 0.45
ensemble (VM93- inspired parameters)	$k_0 = 12$, $\sigma = 1$, $rb = 2$	256 ²	shallow 1–6, none 1–3, deep 1	60	shallow 0.012 $\pm$ 0.003; none 0.06 $\pm$ 0.03

A • Surface stress on WC25 decaying turbulence

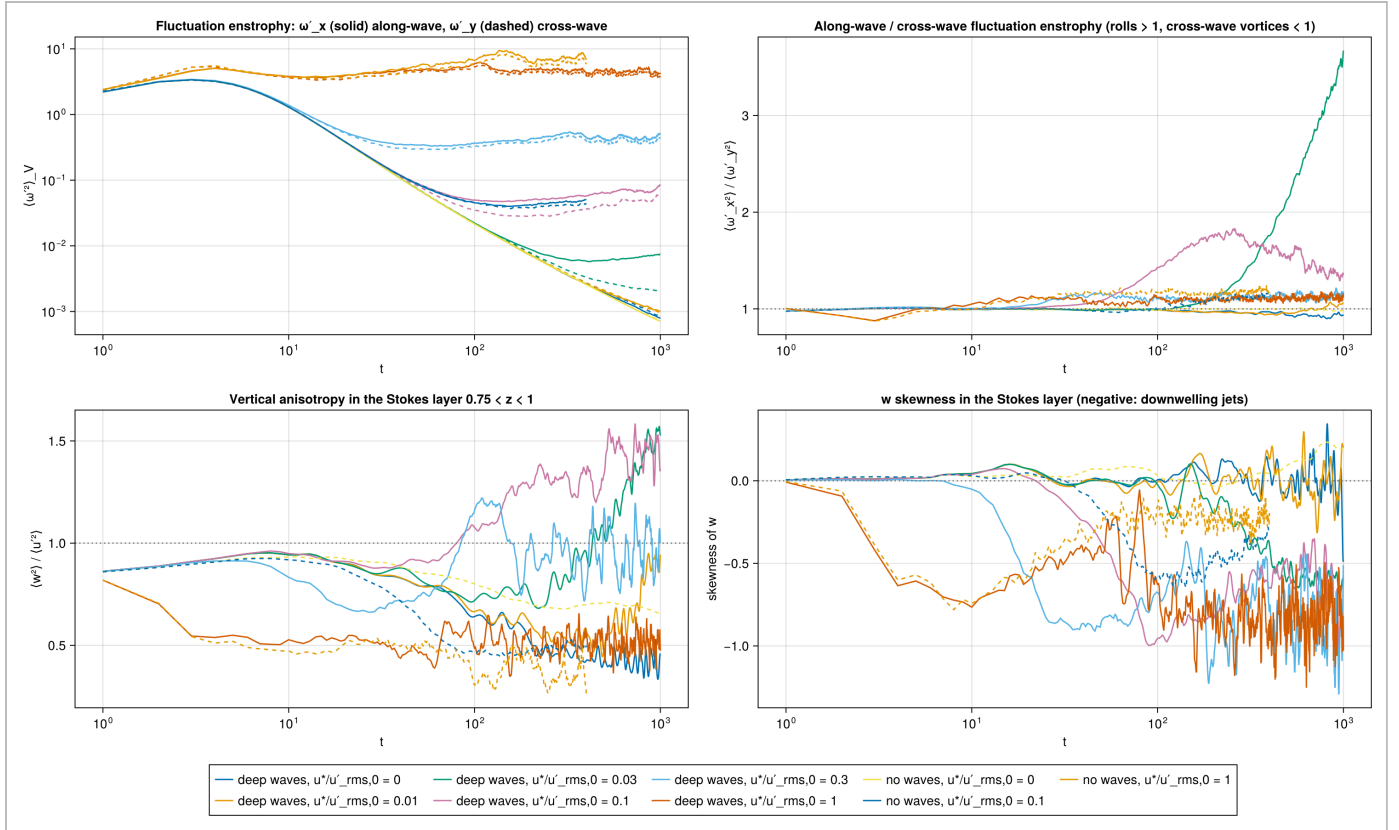
Deep-wave Stokes drift, stress applied along the waves at the surface, decaying turbulence from a shear-free Lagrangian field. Because the fluctuation velocity $u'_{rms}(t)$ decays while τ is fixed, $u^*/u'_{rms}(t)$ increased in every stressed run over the durations sampled, and it is this instantaneous ratio rather than $u^*/u'_{rms,0}$ that organises the cases. Whether a weak stress eventually crosses the roll threshold is not established by these finite-time runs: the stress itself sustains fluctuations (the enstrophy plateaus), so u'_{rms} need not decay indefinitely.



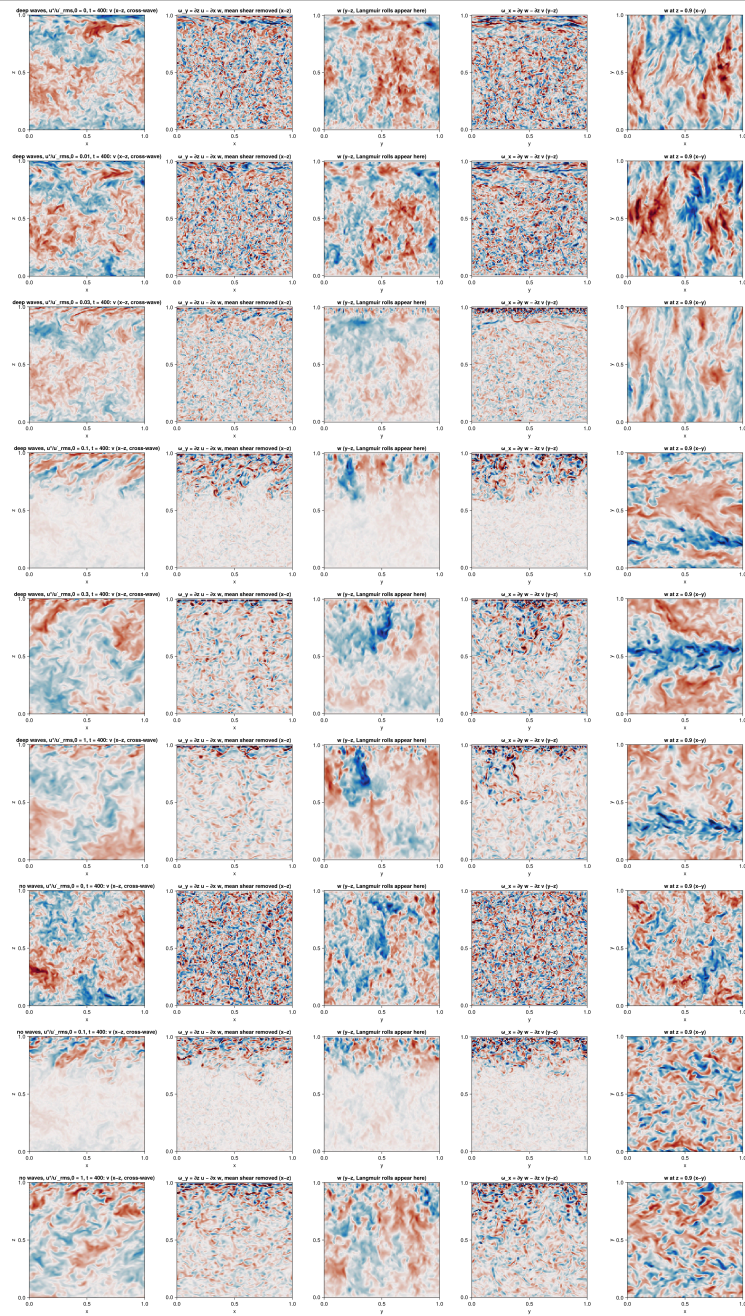
Regime diagram (128³, seed 1). Left: $u^*/u'_{rms}(t)$ ($u'_{rms}(t) = \sqrt{2K_{fluct}}$, the rms of the velocity fluctuation about the horizontal-mean profile) for every stressed case; the ratio grows as the turbulence decays and saturates once the wind layer sets u'_{rms} . Middle: the along/cross fluctuation-ensrophy ratio $\langle \omega'^2_x \rangle / \langle \omega'^2_y \rangle$ against u^*/u'_{rms} ; in the runs shown, the roll signature (ratio > 1.3) appears only along the $u^*/u'_{rms,0} = 0.03$ (La 0.17) and 0.1 (La 0.31) curves, beyond $u^*/u'_{rms} \approx 0.7-1$, reaching 1.75 and 3.7; $u^*/u'_{rms,0} = 0.3, 1$ and both wave-free controls stay below 1.2 at any ratio. Right: Stokes-layer vertical anisotropy $\langle w^2 \rangle / \langle u'^2 \rangle$ rises above 1 (to 1.3–1.5) in the same two cases and falls to ≈ 0.5 for the shear-dominated cases. Figure `final_regime.png`.



Energy budget. Total Lagrangian kinetic energy K/K_0 decays to $1.3e-3$ at $t = 400$ with waves against $0.9e-3$ without (top left); mean-profile energy K_{jet} (top right) grows at zero stress from $t \approx 100$ while the control's falls, and the fraction K_{jet}/K_{rel} (bottom left) reaches 0.40 at $t = 1000$ (control ≤ 0.02). The bulk velocity follows τt exactly (bottom middle, momentum check), and the budget residual $\epsilon = \tau(u)_{top} - dK/dt$ is positive at all times (implicit dissipation only removes energy). Figure `final_energy.png`.



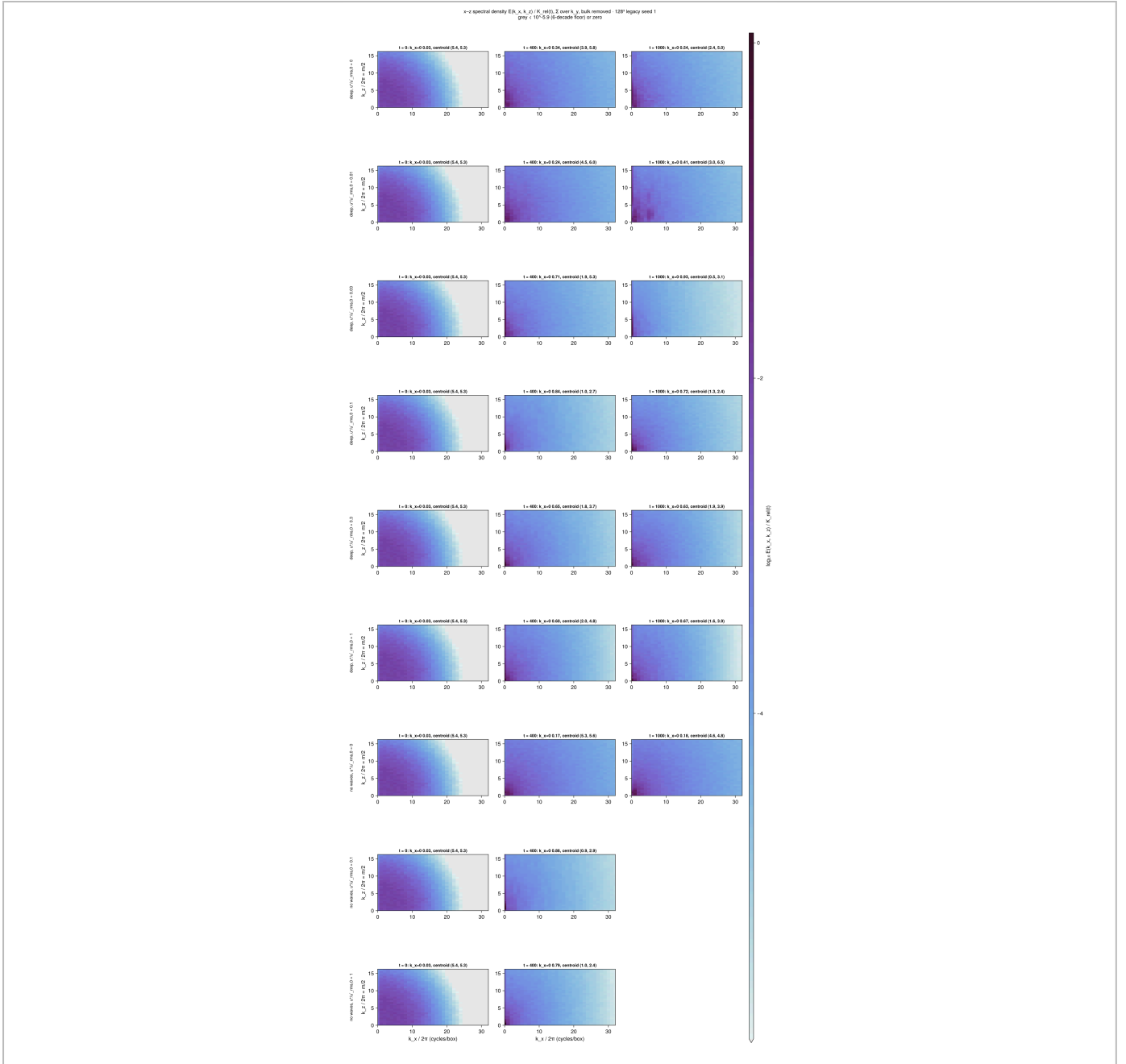
Structure diagnostics against time. Fluctuation enstrophy (top left) decays as $t^{-1.6}$ without stress and plateaus at a stress-set level otherwise ($r = 0.03$ from $t \approx 300$, 0.1 from ≈ 100 , 0.3 from ≈ 30 , 1 from ≈ 5). The roll ratio (top right) departs from 1 at $t \approx 80$ for $r = 0.1$ and $t \approx 277$ for $r = 0.03$ and keeps rising to 3.7 by $t = 1000$ in the latter; the w skewness turns negative (downwelling plumes) at the same times. Figure `final_anisotropy.png`.



Sections at $t = 400$ (rows: deep waves $r = 0, 0.01, 0.03, 0.1, 0.3, 1$; wave-free $r = 0, 0.1, 1$; columns: cross-wave v in x - z , ω_y with the mean shear removed, w in y - z , ω_x , and w at $z = 0.9$). At $r = 0$ and 0.01 , w at $z = 0.9$ is organised into cross-wave (y -elongated) streaks and the y - z vorticity shows no surface-attached cells; at $r = 0.1$ the y - z plane shows surface-attached counter-rotating cells with narrow downwelling plumes reaching $z \approx 0.6$ and w at $z = 0.9$ is x -elongated (Langmuir rolls); at $r = 0.3$ and 1 the wind layer fills the column and the structures match the wave-free controls at the same stress. Figure `final_sections_t400.png`.



Jet metrics. Interior sign reversals of $U_L - U_{\text{bulk}}$ settle at 3 for $r = 0$ and 0.01 by $t \approx 300$ (no reversals for $r \geq 0.1$ once the wind layer forms), the band wavelength reaches ≈ 0.75 at $t = 1000$, the same order as the local Rhines estimate $2\pi\sqrt{(U/|\partial z z u^*|)}$ (dashed; 0.18 at $z = 0.9$, 0.32 at $z = 0.75$) without being a quantitative match, and the relative profile becomes persistent (10-unit autocorrelation ≈ 1) after $t \approx 50$. Figure `final_jets.png`.



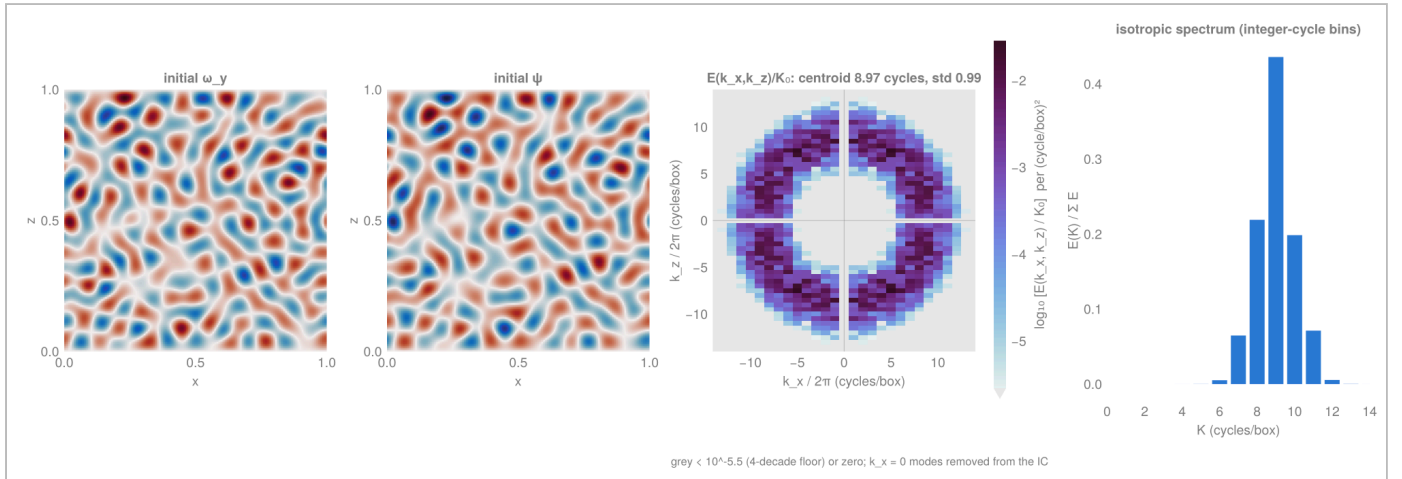
Directional spectra in the x–z plane. $E(k_x, k_z)/K_{rel}(t)$ of the velocity fluctuation about the volume mean, summed over k_y (Fourier in x , cosine/sine in the bounded z direction), at $t = 0, 400$ and 1000 for the 128^3 seed-1 deep-wave sweep and the wave-free controls (the legacy 128^3 draw, a different realization from the 256^3 runs). One cell per mode on axes in cycles per box ($k_z/2\pi = m/2$), equal aspect; $\log_{10}$ colour shared by all panels, six decades, grey below the floor. Energy collects in $k_x = 0$ modes as the mean profile organises: fraction 0.34 ($t = 400$) and 0.54 ($t = 1000$) at zero stress against 0.17–0.18 without waves; $0.71 \rightarrow 0.93$ at $u^*/u'_{rms,0} = 0.03$; 0.84 at 0.1 ($t = 400$). The spectra sum exactly to the relative kinetic energy. The earlier panels in `final_spectra.png` used a DCT/DST scaling that summed to half the energy (fractions and centroids were unaffected). Figure `final_xz_spectra_heatmap.png`.

Empirical brackets from a sampled sweep, not a universal transition. The observed pattern — rolls for $u^*/u'_{rms}(t) \gtrsim 0.7$ –1 with $La \lesssim 0.3$, persisting WC25 jets while $u^*/u'_{rms} \lesssim 0.3$, shear turbulence for $La \gtrsim 0.5$ — rests on one seed at $r = 0.01, 0.1$ and 0.3, two seeds at 0, 0.03 and 1, and two grids at 0, 0.03 and 0.1. It is a description of these runs, not a necessary or sufficient condition. The onset time differs between grids ($r = 0.1$: $t = 82$ at 128^3 , 156 at 256^3) while the onset ratio agrees (1.02 vs 1.06); agreement between two grids is not a demonstration of convergence. *Corrected 2026-10-03*: an earlier version said the $r = 0.03$ onset was less certain at 256^3 because a layer-restricted measure gave $t \approx 270$ while the “raw volume ratio $\langle \omega_x'^2 \rangle / \langle \omega_y'^2 \rangle$ ” was only ≈ 1.2 at $t = 400$ against 1.75 at 128^3 . That compared different quantities: the 1.2 (256^3) is the raw ratio of total vorticity, R_{raw} , which includes the mean shear (128^3 : 1.00), whereas the 1.75 (128^3) is the fluctuation ratio R_{fl} (256^3 : 1.66), and $t \approx 270$ is the R_{fl} onset (definitions under Editorial corrections). With the same measure on both

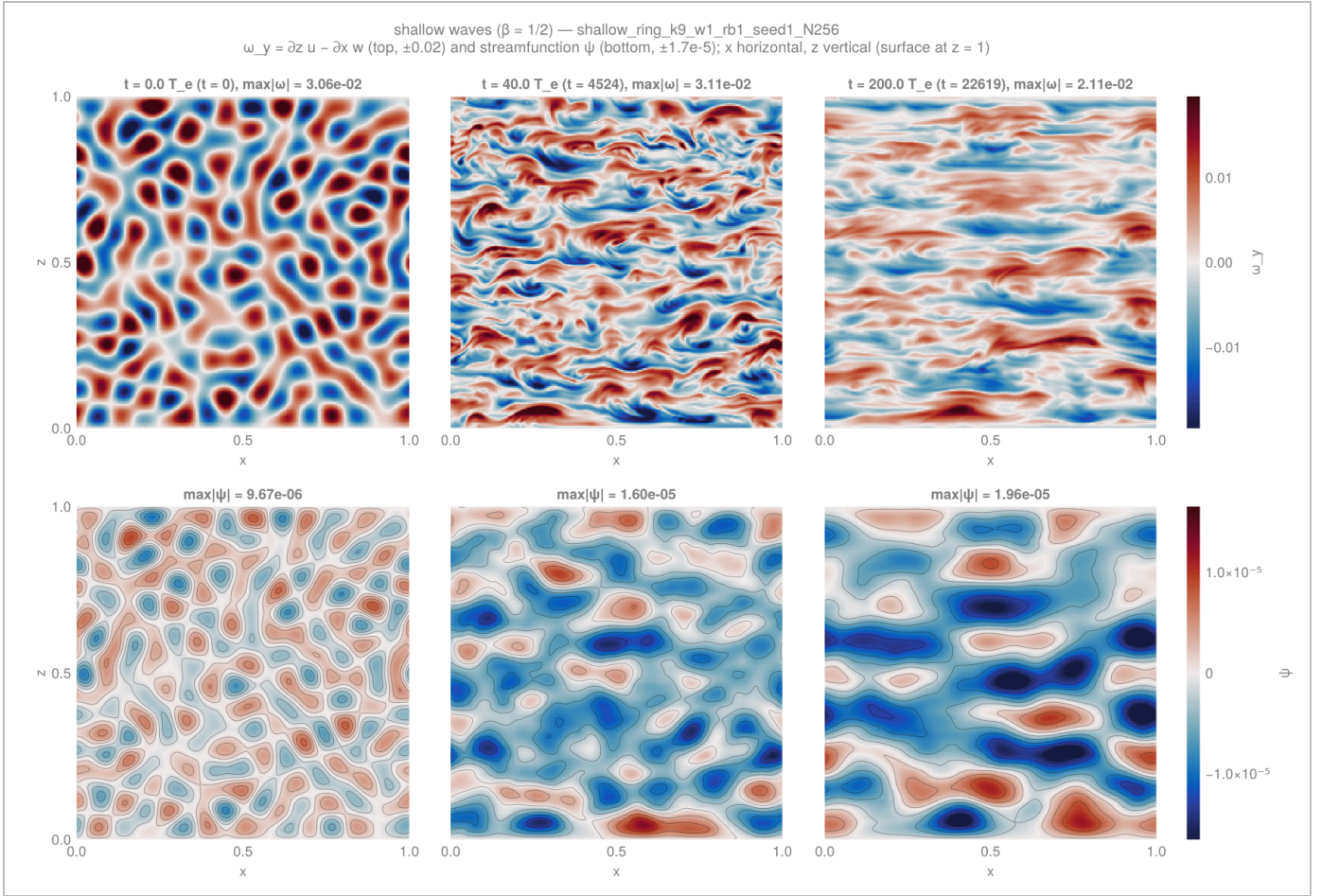
grids the $r = 0.03$ onset agrees (R_{ff} : $t = 277$ and 270). The 128^3 and 256^3 runs compared here started from different realizations; in the interim same-field comparison (REPORT_resolution.md) the $r = 0.1$ onset is $t = 84$ (128^3) and 156 (256^3), so that shift is a resolution effect; 512^3 is pending. The 128^3 sweep is strongly dissipative ($K/K_0 = 2.7\text{e-}3$ by $t = 190$), the jets are 3–10 $\times$ weaker than WC25's spun-up cases, and the box holds only 2–3 bands. At $r \geq 0.3$ the mean advection reaches $O(1)$ and the implicit WENO dissipation becomes velocity-dependent (a Galilean-invariance caveat for the LES, not a CFL-stability problem: the adaptive step kept CFL at 0.5). The 256^3 extension to $t = 1000$ at $r = 0.03$ is now complete (see the interim report). The 512^3 checks, additional stress values and seeds, and domain-sensitivity tests remain relevant before stating quantitative thresholds.

B • Vallis narrow-ring decay in two-dimensional WC25 flow

One Lagrangian ring field ($k_0 = 9$ cycles/box, $\sigma =$ one mode spacing, $rb = 1$, seed 1) evolved under shallow waves, deep waves and no waves; the ring parameters follow the Vallis demo code (its amplitude law $1/[K^2 + (K^2 - 3200)^2]$ peaks at 9.0 cycles and, once its unnormalised FFT is undone, corresponds to $rb \approx 0.96$) and VM93 ($\beta = 400$, $E = 1$, $K = 12$ give $rb \approx 1.96$ under the assumed source normalization; their $Z/E = 2K^2$ is consistent with a narrow ring but does not determine its width, so $\sigma = 1$ mode is a chosen value tested against $\sigma = 0.5$ and 2). The comparison with the published figures is qualitative and source-matched, not an exact reproduction.

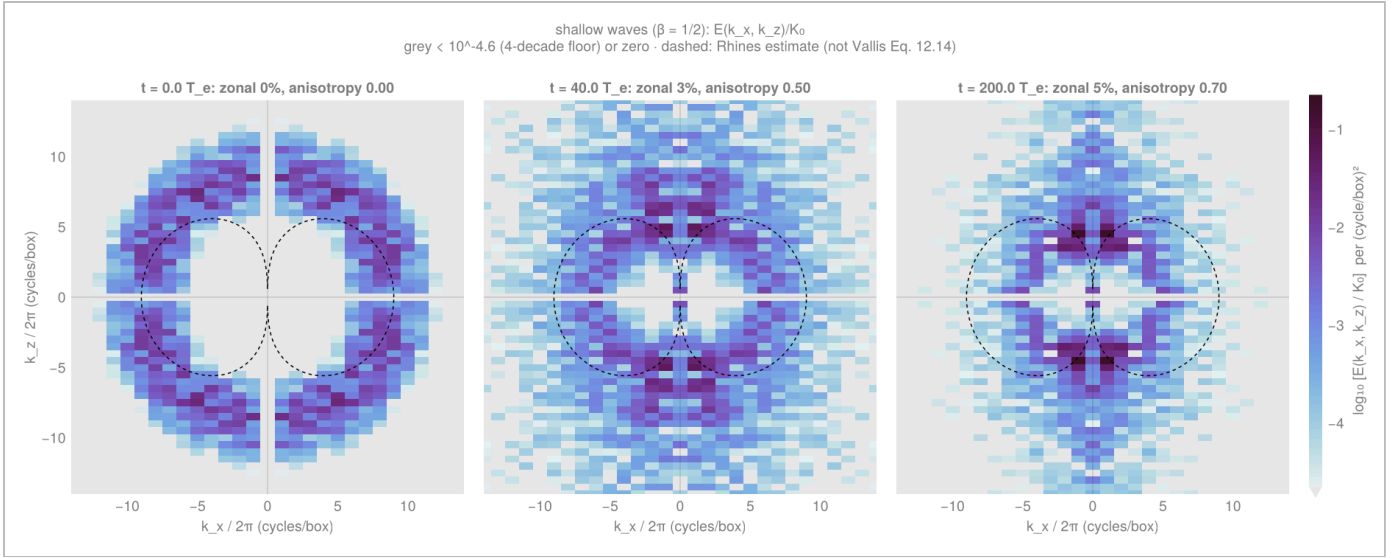


Shared initial condition (256^3): ω_y , ψ , $E(k_x, k_z)$ and the isotropic spectrum. Centroid 8.97 cycles, standard deviation 0.99, divergence $7\text{e-}18$, $w = \omega_y = 0$ on the walls, anisotropy index 0.004 (0.057 if the $k_x = 0$ modes were kept). $u'_{\text{rms},0} = 1.56\text{e-}4$, $K = 1.22\text{e-}8$, enstrophy $7.84\text{e-}5$. Figure final_ic_heatmap.png. Heatmap: spectral density per (cycle/box) 2 , normalized by the initial energy K_0 , one cell per lattice mode ($k_x/2\pi$ integer, $k_z/2\pi = n/2$ for the wall sine basis), $\log_{10}$ colour on a perceptually uniform sequential map with limits shared across the panels and families being compared; cells more than four decades below the shared maximum, or exactly zero, are grey. No interpolation. The earlier filled-contour version is kept as final_ic.png.

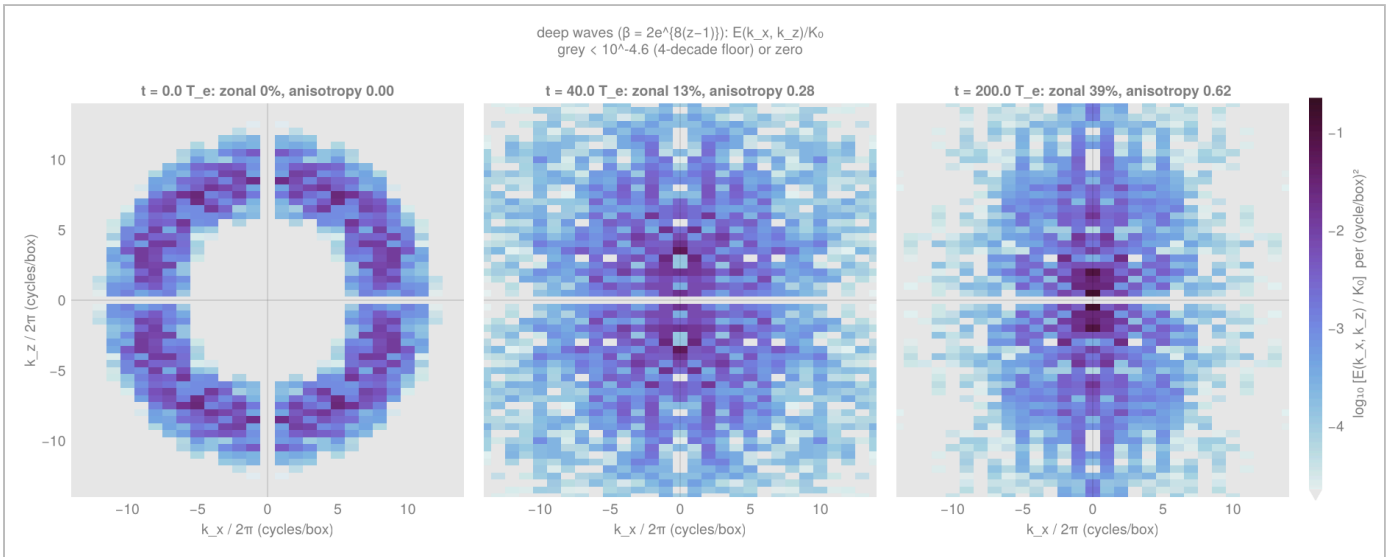


Shallow waves ($\beta = 1/2$), the Fig. 12.4 analogue. ω_y (top, fixed limits ± 0.02) and ψ (bottom, $\pm 1.7e-5$) at 0, 40 and 200 T_e . The isotropic blob pattern becomes x-elongated streaks by 40 T_e and bands by 200 T_e ; ψ organises into x-elongated cells with 4–5 alternations across the depth. Energy is conserved to +0.03%, enstrophy decays to 39%. Figure `final_panels_shallow.png`.

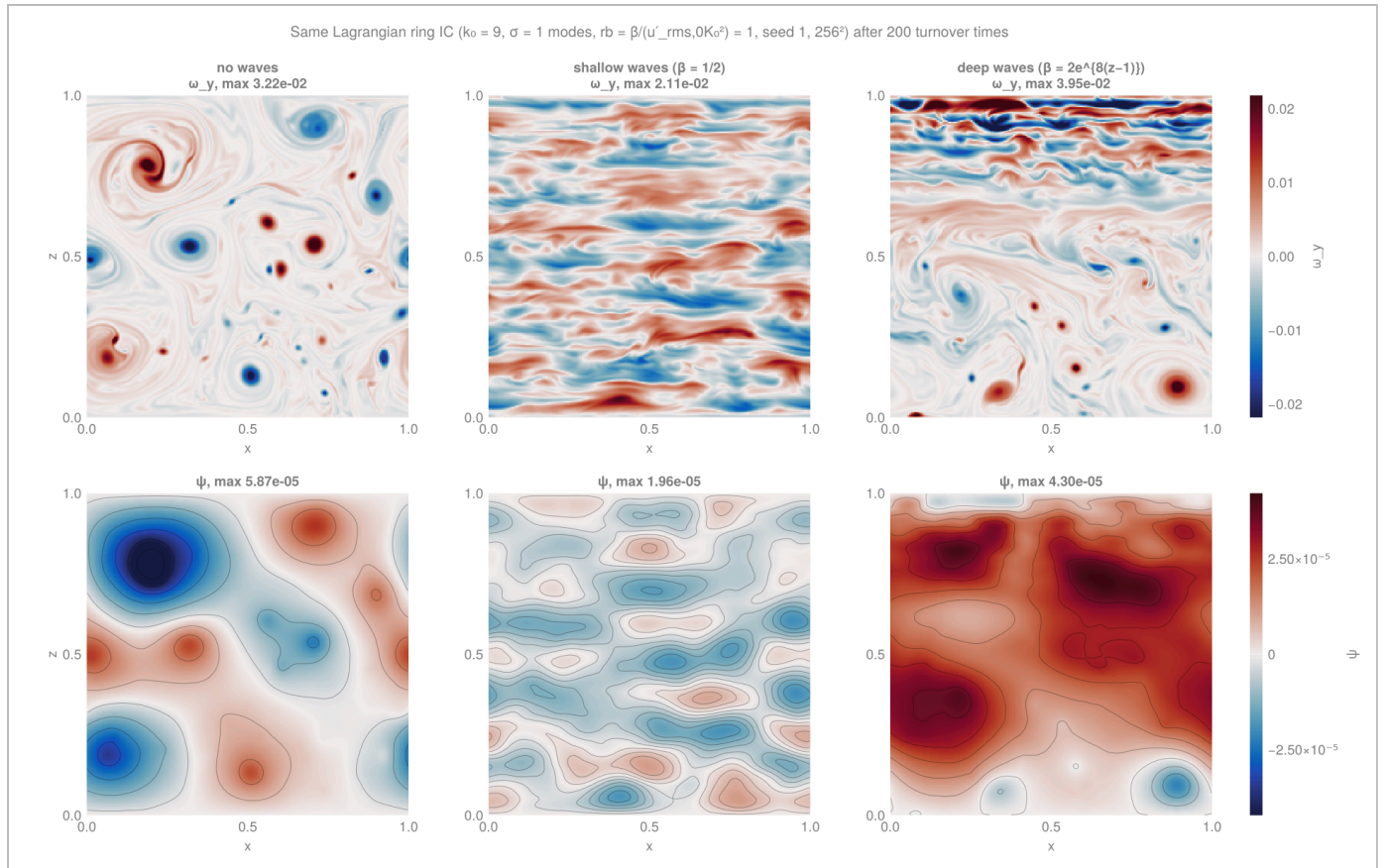
Overlay audit (2026-10-04). The dashed curves use $K = \sqrt{(\beta |\cos \theta| / u'_{rms})}$ with the correct conversion $K/(2\pi)$ to cycles/box. They are bulk-velocity scale estimates, not fitted boundaries of the observed spectral holes. Vallis Eq. 12.14 instead uses $K = (\beta^3/\epsilon)^{1/5} |\cos \theta|^{3/5}$, requiring a defensible inverse energy transfer rate ϵ . A flux-based audit is pending; no curve has been resized to fit the data.



Shallow waves, the Fig. 12.3 analogue. $E(k_x, k_z)/K_0$ at 0, 40 and 200 T_e with a Rhines-scale estimate $K^2 = \beta |\cos \theta| / u'_{\text{rms}}(t)$ dashed (a velocity-based estimate using the instantaneous rms speed, not the textbook ε -based boundary of Vallis Eq. 12.14, whose shape differs). The energy leaves the ring and becomes concentrated near the k_z axis; the dashed curve is too large to trace the observed depleted region. It remains near a maximum radius of 9 cycles/box as total rms speed changes little. The spectral concentrations lie ($|k_x| \approx 3$, $k_z \approx 4-8$ cycles at 40 T_e; $|k_x| \leq 2$, $k_z \approx \pm 4-5$ at 200 T_e). Anisotropy index 0.00 $\rightarrow$ 0.50 $\rightarrow$ 0.70; exact-zonal fraction 0 $\rightarrow$ 3% $\rightarrow$ 5%. Figure `final_spectra_shallow_heatmap.png`. Heatmap: spectral density per (cycle/box)², normalized by the initial energy K_0 , one cell per lattice mode ($k_x/2\pi$ integer, $k_z/2\pi = n/2$ for the wall sine basis), log₁₀ colour on a perceptually uniform sequential map with limits shared across the panels and families being compared; cells more than four decades below the shared maximum, or exactly zero, are grey. No interpolation. The earlier filled-contour version is kept as `final_spectra_shallow.png`.



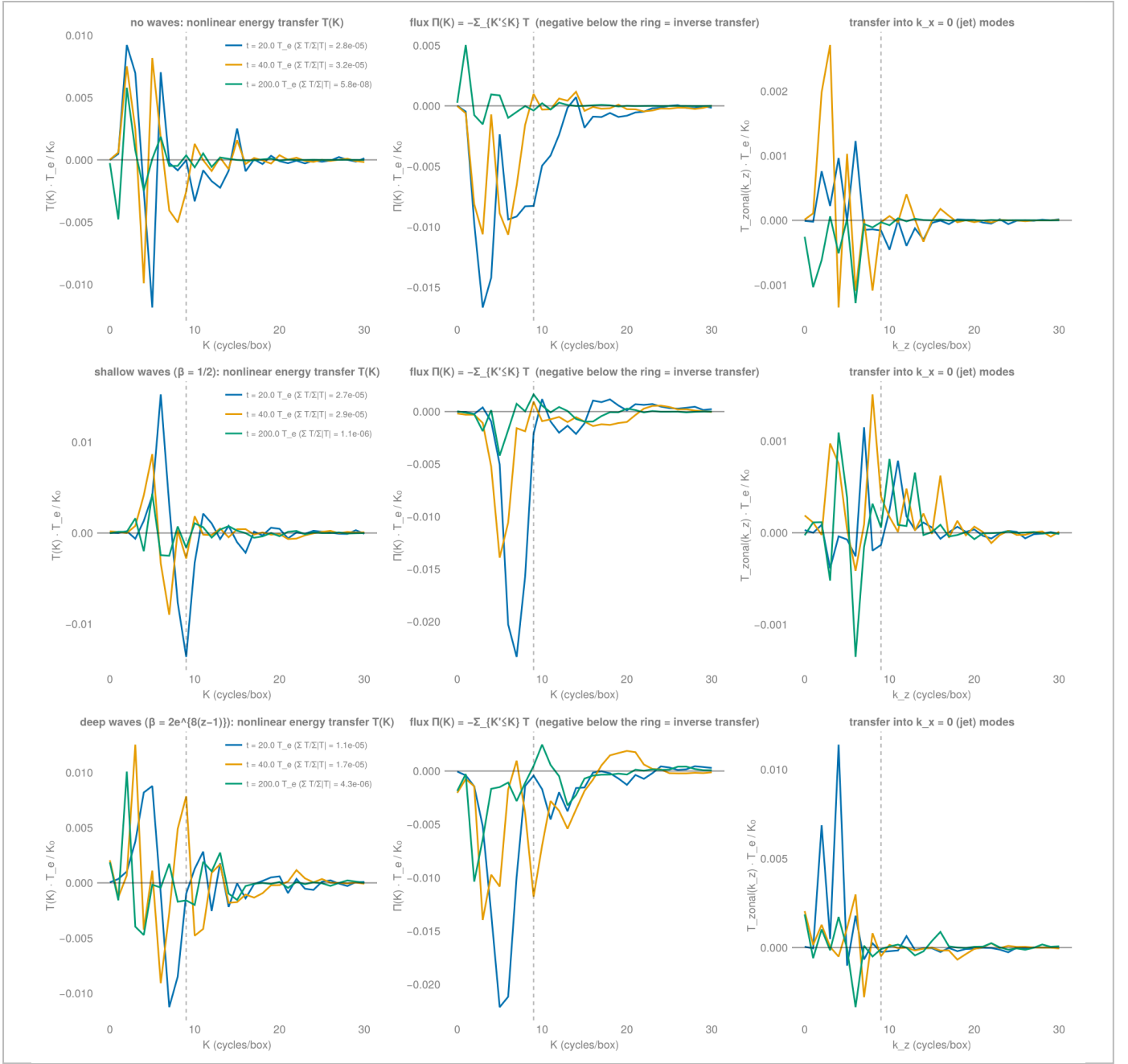
Deep waves, same diagnostic. $E(k_x, k_z)/K_0$ for the deep-wave member (same ring initial condition, seed 1, 256²) at 0, 40 and 200 T_e, on the same colour scale as the shallow-wave figure above: log₁₀ limits -4.65 to -0.65 , shared by the shallow, deep and wave-free spectra, grey below the four-decade floor or exactly zero, so the two figures compare cell by cell. The spectrum is a Fourier (x) $\times$ sine (z) decomposition of the streamfunction over the whole column between the walls: $k_z/2\pi = n/2$ counts vertical half-wavelengths across the domain, the $k_z = 0$ row is empty because the sine basis has no $n = 0$ mode, and a given (k_x, k_z) cell mixes contributions from every depth rather than describing waves at one level. No dumbbell is drawn: $\beta_{\text{eff}}(z) = \partial z u^* = 2e^{\{8(z-1)\}}$ varies by a factor of about 3000 over the depth, so a constant- β Rhines estimate has no single value here. Energy leaves the ring and concentrates in and next to the exact-zonal $k_x = 0$ column at low $|k_z|$ (about 0.5–2 cycles per box): exact-zonal fraction 0 $\rightarrow$ 13% $\rightarrow$ 39% and anisotropy 0.00 $\rightarrow$ 0.28 $\rightarrow$ 0.62, whereas with shallow waves the energy collects at the dumbbell poles off the $k_x = 0$ axis (exact-zonal 5%, anisotropy 0.70 at 200 T_e). Single realization at one grid; the deep-wave jet statistics are seed-sensitive and change by roughly 10–20% across the higher-resolution comparisons (exact-zonal fraction 0.38 at 512² and 0.45 at 1024² at 200 T_e; see `REPORT_resolution.md`). Figure `final_spectra_deep_heatmap.png` (`analysis/wc25_vallis2d/figures.jl`, `spectra_only=true`, the same render as the shallow figure).



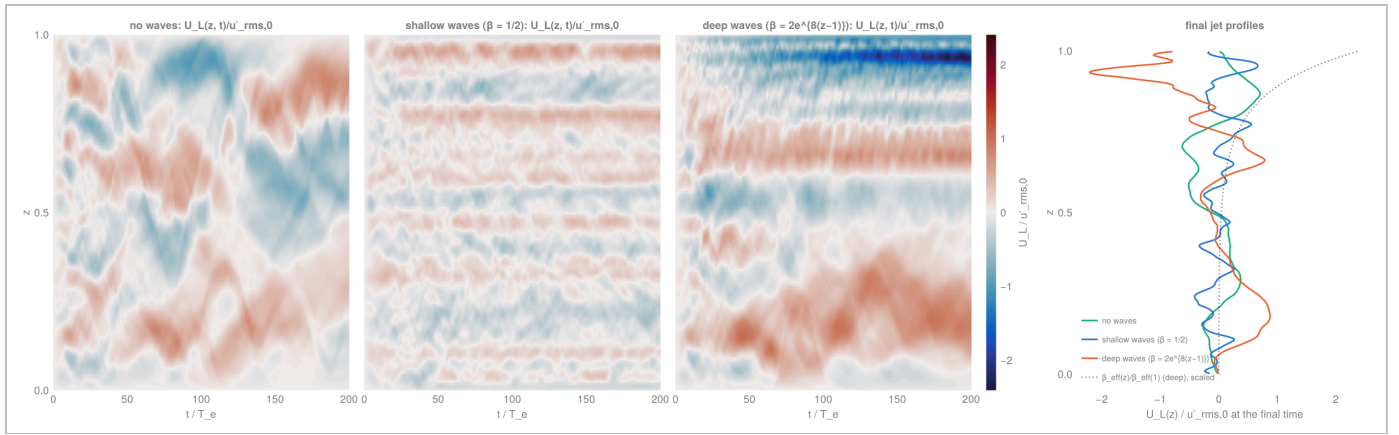
Three families at 200 T_e (ω_y top, ψ bottom, shared limits). No waves: Vallis Fig. 11.8 decay — coherent vortices, merger, a flat enstrophy landscape, energy centroid 2.6 cycles, enstrophy 14%. Shallow: x-elongated bands, centroid 5.1 cycles, enstrophy 39%. Deep: β dynamics in the upper third (where local $\beta/(u'_{rms,0}K_0^2)$ exceeds ≈ 0.1), 2-D vortices below $z \approx 0.6$, and a mean streamfunction of one sign carrying the x-mean jet. Figure final_comparison_t200.png.



Time series. K/K_0 stays within +0.03% (shallow), -0.6% (none) and $+0.6\%$ (deep; see caveats) while the β -plane reference loses 1% to hyperviscosity. Enstrophy falls to 39–42% with waves and 14% without. K_{jet}/K reaches 0.05 (shallow), 0.12 (none, box-scale condensate) and 0.39 (deep); $\max|U_L|/u'_{rms,0}$ reaches 0.7, 0.7 and 2.2. The anisotropy index rises to 0.70 (shallow) and 0.62 (deep) and stays ≈ 0 without waves. The energy centroid falls from 9 to 5.1 (shallow), 4.1 (deep) and 2.6 cycles (none). Figure final_timeseries.png.



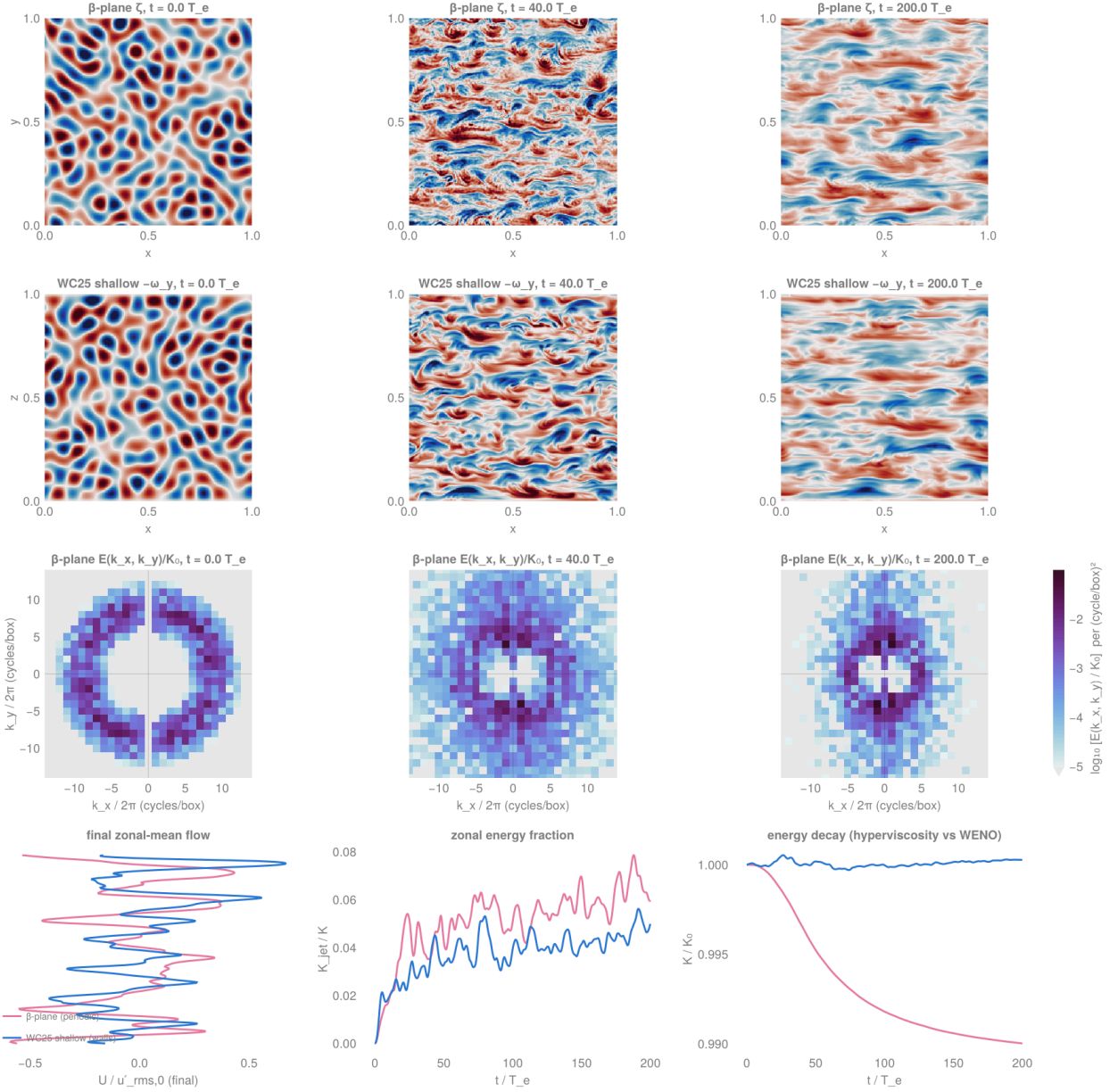
Spectral transfer. $T(K)$ is negative at the ring ($K \approx 7-9$) and positive at $K \approx 4-6$, so the flux $\Pi(K) = -\sum_{K' \leq K} T$ is negative below the ring: an inverse energy transfer, strongest at $20 T_e$ and weakening by $200 T_e$ ($\sum T/\sum |T| \approx 1e-5$, so the diagnostic is conservative). The transfer into $k_x = 0$ modes is $10\times$ larger with deep waves than with shallow waves and concentrated at $k_z \approx 2-4$ cycles. Figure `final_transfer.png`.



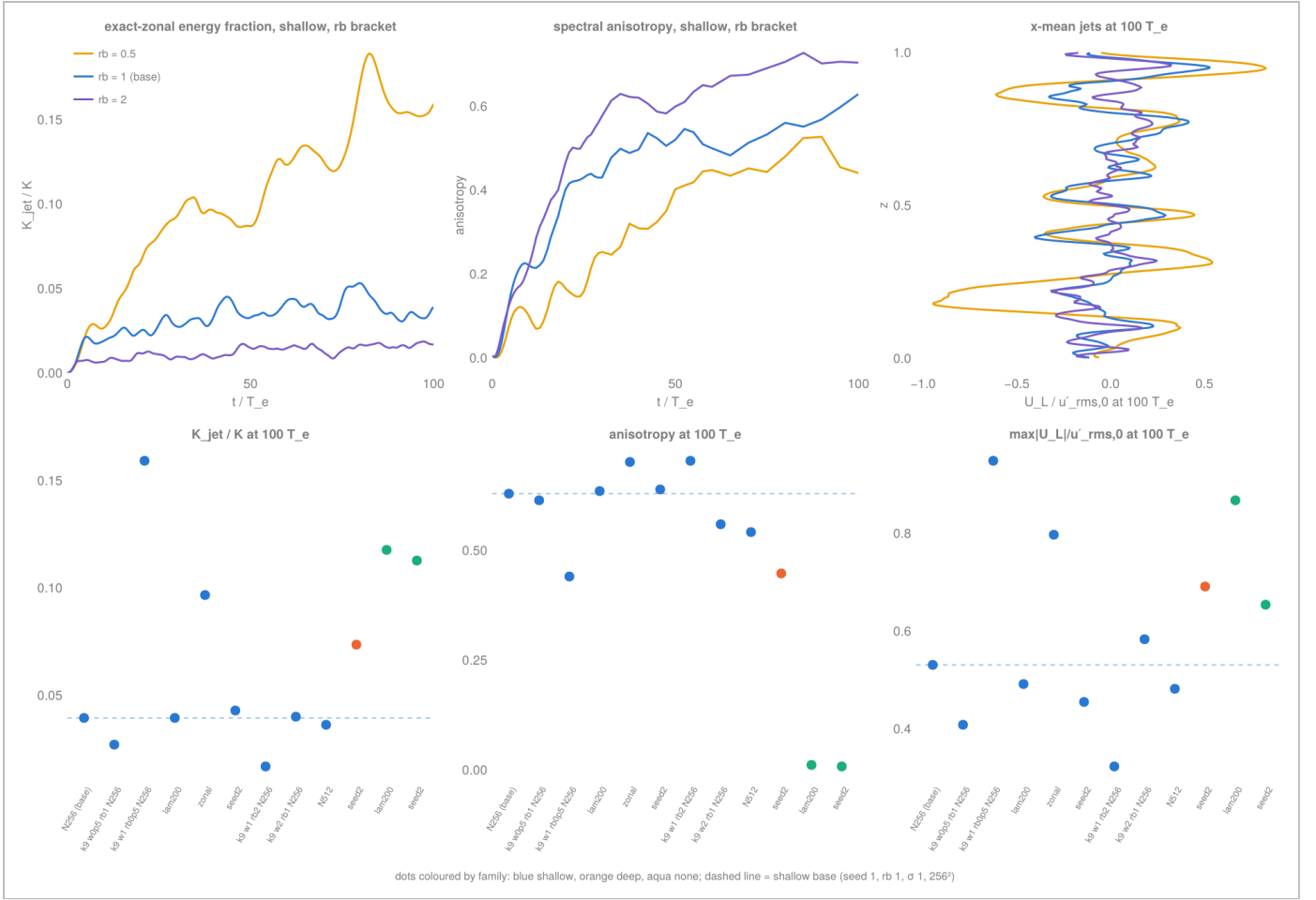
x-mean jets. $U_L(z, t)/u'_{rms,0}$ for no waves (left), shallow waves (middle) and deep waves (third), with final profiles at right. Shallow: many fine, weak bands (max $0.67 u'_{rms,0}$, wavelength ≈ 0.09). Deep: a surface jet toward $-x$ at $z \approx 0.9$ reaching $-2.2 u'_{rms,0}$ and interior jets toward $+x$ near $z \approx 0.65$ and 0.2 (4 sign reversals, wavelength ≈ 0.2), growing almost linearly in K_{jet}/K to 0.39 . Figure `final_jets.png`.

Doubly periodic β -plane reference ($\beta = 1/2$) vs WC25 shallow waves with walls

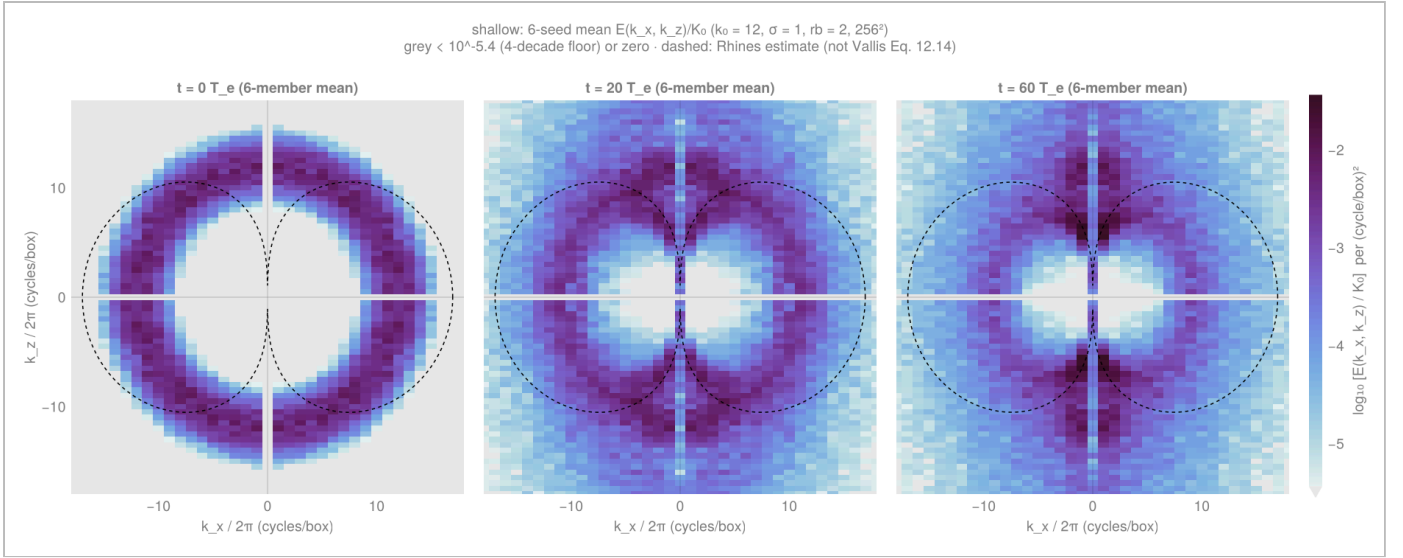
row 3: periodic reference $E(k_x, k_y)/K_0$, integer cells - grey < $10^{-5.0}$ (4-decade floor) or zero



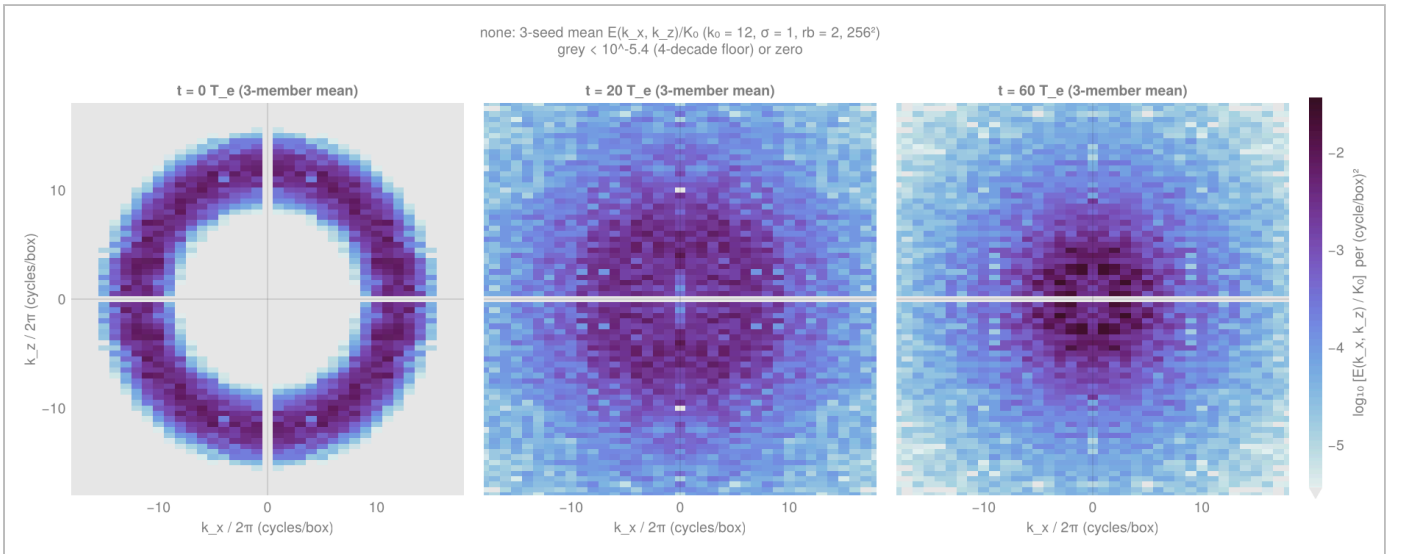
Textbook geometry against walls. The doubly periodic pseudo-spectral $\beta = 1/2$ reference (same ring law, same $u'_{rms,0}$, RK4, $2/3$ dealiasing, ∇^8 hyperviscosity) and the WC25 shallow case with walls give the same vorticity streaks, the same dumbbell spectra and the same exact-zonal fraction (0.060 vs 0.050); the walls are not what limits the jets at $rb = 1$. Energy: -1% (hyperviscosity) vs $+0.03\%$ (WENO). Figure `final_betaplane_heatmap.png`. Heatmap: spectral density per $(\text{cycle/box})^2$, normalized by the initial energy K_0 , one cell per lattice mode. Lattices differ between the two geometries: the doubly periodic reference (unit box, Fourier in both directions) has integer spacing in both $k_x/2\pi$ and $k_y/2\pi$, so its cells are 1×1 $(\text{cycle/box})^2$; the WC25 wall case uses a sine basis in z with $k_z/2\pi = n/2$, i.e. half-integer spacing and $1/2$ $(\text{cycle/box})^2$ cells, so its spectra (other figures) show twice as many rows in k_z , $\log_{10}$ colour on a perceptually uniform sequential map with limits shared across the panels and families being compared; cells more than four decades below the shared maximum, or exactly zero, are grey. No interpolation. The earlier filled-contour version is kept as `final_betaplane.png`.



Sensitivity at 100 T_e. Top: the rb bracket for shallow waves — exact-zonal fraction 0.159 (rb = 0.5), ≈ 0.04 (rb = 1), 0.017 (rb = 2) while the anisotropy index runs the other way (0.44, 0.62, 0.70): a stronger β arrests the cascade nearer the ring and keeps the energy in zonally elongated waves rather than jets. Bottom: every variant at 100 T_e, coloured by family — ring width 0.5–2 modes, 512², seed 2 and the $\lambda = 200$ rescaling ($u'_{rms,0} = 0.031$, identical physics in the nonlinear WENO regime: 0.039 vs ≈ 0.04 , with 1% energy loss instead of +0.03%) all sit within the scatter; keeping the $k_x = 0$ modes adds their 6% almost additively (0.097); deep seed 2 gives 0.074 against ≈ 0.2 for seed 1. Figure final_variants.png.



VM93 Fig. 5 / Vallis Fig. 12.3 analogue. Six-seed ensemble mean $E(k_x, k_z)$ for shallow waves with VM93-inspired parameters ($k_0 = 12, \sigma = 1, rb = 2$) at 0, 20 and 60 T_e (VM93's panels sit at $\approx 0, 17$ and 50 turnovers). The energy leaves the ring, develops depleted regions near the k_x axis and forms a narrow hourglass along k_z by 60 T_e . The dashed global Rhines estimate has a maximum radius near 17 cycles/box and is larger than the observed depleted region; it does not quantitatively delimit it; the dashed curve is the velocity-based Rhines estimate, not the ε -based Eq. 12.14 contour. Exact-zonal fraction 0.012 ± 0.003 — small, as VM93 noted for the wave regime. Figure `final_ensemble_spectra_shallow_heatmap.png`. Heatmap: spectral density per $(\text{cycle/box})^2$, normalized by the initial energy K_0 , one cell per lattice mode ($k_x/2\pi$ integer, $k_z/2\pi = n/2$ for the wall sine basis), $\log_{10}$ colour on a perceptually uniform sequential map with limits shared across the panels and families being compared; cells more than four decades below the shared maximum, or exactly zero, are grey. No interpolation. The earlier filled-contour version is kept as `final_ensemble_spectra_shallow.png`.



Wave-free ensemble mean (three seeds, same ring): isotropic collapse toward small K with no preferred direction; exact-zonal fraction 0.06 ± 0.03 from the condensate's random projection onto the x-mean. Figure `final_ensemble_spectra_none_heatmap.png`. Heatmap: spectral density per $(\text{cycle/box})^2$, normalized by the initial energy K_0 , one cell per lattice mode ($k_x/2\pi$ integer, $k_z/2\pi = n/2$ for the wall sine basis), $\log_{10}$ colour on a perceptually uniform sequential map with limits shared across the panels and families being compared; cells more than four decades below the shared maximum, or exactly zero, are grey. No interpolation. The earlier filled-contour version is kept as `final_ensemble_spectra_none.png`.

Controls, resolution and seed dependence

CHECK	A · SURFACE STRESS	B · VALLIS-2D
Controls	Wave-free runs from the same Lagrangian field at $r = 0, 0.1, 1$: bands below the fluctuation rms at $r = 0$ ($K_{\text{jet}}/K_{\text{rel}} \leq 0.02$); roll anisotropy 1.17–1.24 at $r = 0.1, 1$ — the same as the wave cases at $La \geq 0.5$. Wave-only quiescent state exactly at rest.	Wave-free member from the identical field: Fig. 11.8 vortex decay, anisotropy ≈ 0 , remaining enstrophy 14% versus 39–42% with waves (86% lost versus 58–61%; the remaining fractions differ by about threefold, not the losses). Doubly periodic β -plane reference matches the wall case. Quiescent wave-only state exactly at rest.
Resolution	256 ³ at $r = 0, 0.03, 0.1$: surface jet $-3.21\text{e-}3$ vs $-3.24\text{e-}3$ and interior amplitude $8.4\text{e-}4$ vs $7\text{e-}4$ at $r = 0$ ($K_{\text{jet}}/K_{\text{rel}}$ 30% lower); $r = 0.1$ matches 128 ³ within 10% in u'_{rms} , u' , surface mean and K ; onset time at $r = 0.1$ shifts $82 \rightarrow 156$ while the onset ratio stays 1.02–1.06. Two grids agree on the ratio; this is consistency between two resolutions, not demonstrated convergence.	512 ² from the same coefficients: K_{jet}/K 0.050 vs 0.050 (shallow), 0.384 vs 0.400 (deep), anisotropy 0.715 vs 0.696, identical $U_L(z)$ up to small-scale detail, 6% more enstrophy at 200 T_e ; deep energy drift $+0.6\% \rightarrow +0.1\%$ over 200 T_e (decreases with resolution; not shown to vanish). Agreement between two grids; a third would be needed to claim convergence.
Seeds	Seed 2 at $r = 0, 0.03, 1$ and control 0: surface jet $-3.5\text{e-}3$ vs $-3.2\text{e-}3$, interior $8.8\text{e-}4$ vs $7\text{e-}4$ (10–20%); same transition behaviour at $r = 0.03$. One seed only at $r = 0.01, 0.1, 0.3$.	Shallow seed 2 matches seed 1 (0.043 vs ≈ 0.04); six-seed ensemble spread ± 0.003 at $k_0 = 12$; deep x-mean jet is realisation-sensitive (0.074 vs ≈ 0.2 at 100 T_e).
Time stepping	CFL 0.5 wizard, max Δt 0.1; restart bitwise identical at fixed Δt .	$\Delta t = 0.25 \dots 4$ converge to $< 1\text{e-}4$ (end-time matched); production $\Delta t = 1$ (0.79 for deep, wave limit); restart bitwise identical.

Resource use

CAMPAIGN	JOBS	BILLED GPU-H	NOTES
A · surface stress	31	10.83	128 ³ sweep + extensions to $t = 1000$ (≈ 7 GPU-h), three 256 ³ cases (1.5), seed 2 (1.0), medium family (0.4), validation (0.9), analysis and animations (0.4)
B · Vallis-2D	12	2.11	validation (0.10), pilots incl. the env-truncation rerun (1.13), 512 ² (0.19), β -plane (0.07), sensitivity (0.24), ensemble (0.23), figure passes (0.15); a 256 ² member costs 1–3 min once output I/O is sparse
total	43	12.94	ceiling 60 GPU-h (Vallis sub-budget 8); at most 4 concurrent single-GPU jobs; partition default memory. Job-level ledger: <code>job_ledger.md</code> .

Caveats

Deep-wave discrete energy drift (Vallis-2D). In the continuum the Stokes terms do no work on u^L , and the shallow and wave-free members conserve energy to ± 0.03 – 0.6% . The deep member gains $+0.6\%$ over $200 T_e$ at 256^2 and $+0.1\%$ at 512^2 : the staggered-grid interpolation of $\partial_z u^s(w \hat{x} - u \hat{z})$ is not exactly energy-neutral when $\partial_z u^s$ varies on the grid scale (e-folding depth $1/8 = 32$ cells at 256^2). The drift is a discretisation residual that decreases with resolution ($+0.6\% \rightarrow +0.1\%$ between the two grids over $200 T_e$; it has not been shown to vanish) and does not alter the qualitative picture, but deep-wave energy budgets at 256^2 should not be read to better than 1% .

Surface-stress thresholds are observations in a sampled setup. See §A: single seeds at most ratios, two grids, strongly dissipative 128³ sweep, weak jets relative to WC25, box-confined bands, velocity-dependent implicit dissipation at $La \geq 0.5$, finite durations. No universal transition is asserted and no claim is made that weak stresses eventually produce rolls; $u^*/u'_{rms} \gtrsim 0.7$ – 1 with $La \lesssim 0.3$ is where rolls were seen in these runs.

- **Not quantitative reproductions.** A: $u'_{rms,0}$ and k_e are pilot choices and WC25's spun-up IC was not reproduced. B: VM93's conventions are inferred ($E = u'_{rms,0}{}^2/2$, $Z = \langle \zeta^2 \rangle$, 2π box) and Vallis' Fig. 11.8 IC is known only as "a few modes around wavenumber 9"; the comparison is the dumbbell shape and zonal elongation, not amplitudes.
- **Finite time.** B's shallow case at $rb = 1$ has not formed finite-amplitude exact-zonal jets by $200 T_e$; the Rhines wavenumber coincides with the ring by construction. A's $r = 1$ to $t \approx 1000$ and the 256³ cases stop at $t = 400$ – 1000 .
- **Numerics.** WENO(9) Z-weights use $\epsilon = 1e-8$; at $u'_{rms,0} \sim 1e-4$ (B) the scheme is effectively a linear 9th-order upwind scheme, tested by the $\lambda = 200$ member (same phenomenology, 1% more dissipation). The pseudo-spectral reference uses ∇^8 hyperviscosity, so its dissipation differs by construction.
- **Walls and symmetry (B).** The sine basis makes the k_z lattice spacing π (mode counts not isotropic) and removes $k_x = 0$ modes; free-slip walls admit boundary-trapped x-mean flow (deep case, $z \approx 0.9$).

- **Sign convention.** WC25 prints $\partial z u^s = \pm z/2$ in different places; both campaigns use $+z/2$, which fixes the Rossby phase direction ($-x$) and nothing else.
- **Operational.** A's first validation attempt and B's first two failed on test-side bugs (fixed, cost 0.8 GPU-h in total); B's first pilot lost its 512² members to whitespace truncation in the submission tool's environment passing (fixed on both sides); all rerun.

Editorial corrections to the source reports

Both campaign reports preserve the original results, with updated velocity notation and dated, bracketed correction notes (REPORT_surface_stress.md, REPORT_vallis2d.md) for traceability. Following an independent review of the surface-stress report, this synthesis departs from the source wording in the following places; these changes correct definitions and interpretation without changing simulation data.

- **Langmuir number at zero stress.** The source lists $La = \infty$ for $u^*/u'_{rms,0} = 0$. With $La = \sqrt{(u^*/u^s(1))}$, zero stress with finite waves gives $La = 0$. For the wave-free controls we label La not applicable: the formula is singular for positive stress and indeterminate at zero stress. The case table above uses 0 and "undefined" accordingly.
- **"Every finite stress eventually crosses the first threshold."** Dropped. The finite-time runs show $u^*/u'_{rms}(t)$ increasing in every stressed case, but the stress also sustains fluctuations (plateauing enstrophy), so an eventual crossing for weak stress is not established.
- **"Regime rule".** Rephrased as the pattern observed in this sampled sweep (one seed at most ratios, two grids at three ratios), not as necessary or sufficient conditions for rolls.
- **"Resolution test of the rule" / "256³ confirms".** Rephrased as agreement between two grids; convergence is not demonstrated by two resolutions.
- **"Exceed CFL-safe velocities".** The high-stress cases are integrated at CFL 0.5 by the adaptive step; the caveat is that $O(1)$ mean advection makes the implicit WENO dissipation velocity-dependent (Galilean-invariance caveat for the LES), which is distinct from a CFL stability problem.
- **"The Lagrangian mean stays nearly shear-free".** Qualified: near-shear-free relative to the Stokes profile ($U_E \approx -u^s$ persists), apart from the surface jet and interior bands that are the subject of the campaign.
- **Final times and onset measures.** The deep $r = 1$ extension was still finishing at $t \approx 1000$ when the report was written ($t = 400$ is the completed common end time). *Corrected 2026-10-03 (classifier audit):* an earlier version of this item attributed the 256³ $r = 0.03$ onset ($t \approx 270$) to a layer-restricted measure and called the timing uncertain because it disagreed with the raw volume ratio (≈ 1.2 at $t = 400$). Both statements were wrong. The $t \approx 270$ onset comes from the depth-mean fluctuation ratio R_{fl} defined below; the raw volume ratio is a different quantity that includes the mean shear and is not a roll classifier. With consistent classifiers the two grids agree: R_{fl} onset $t = 277$ (128³) and 270 (256³), layer-restricted R_{layer} onset $t = 178$ and 167. The legacy 128³ and 256³ initial conditions are different realizations, so these are realization-plus-resolution comparisons (see REPORT_resolution.md).
- **Roll-ratio definitions and the primed raw ratio (added 2026-10-03).** Standard curl signs: $\omega_x = \partial_y w - \partial_z v$ (along-wave axis), $\omega_y = \partial_z u - \partial_x w$ (cross-wave axis). $\mathbf{R}_{raw} = \langle \omega_x^2 \rangle_V / \langle \omega_y^2 \rangle_V$: ratio of volume means of the *total* squared components (scalar statistics), which include the mean-shear terms $(\partial_z V)^2$ and $(\partial_z U)^2$; the wind layer's $(\partial_z U)^2$ keeps it near 1 even when rolls are present. $\mathbf{R}_{fl} = \langle \omega'_x{}^2 \rangle / \langle \omega'_y{}^2 \rangle$ (the roll classifier for all onset times and the ratio plotted in the anisotropy and regime figures): horizontal means of ω_x^2 , ω_y^2 on the $N + 1$ z -faces, minus $(\partial_z V)^2$ and $(\partial_z U)^2$ from the mean profiles (an exact removal of the mean-shear part), averaged with equal weights over all faces, then divided. $\mathbf{R}_{layer}$: the same with the depth mean restricted to faces with $z > 0.75$. Onset: first $t > 5$ with the ratio > 1.3 . The source report's sentence "a slightly earlier roll onset ($\langle \omega_x^2 \rangle / \langle \omega_y^2 \rangle \approx 1.2$ at $t = 400$ in the raw measure)" for 256³ $r = 0.03$ primes a raw quantity: the 1.2 is R_{raw} (1.175 at

256³ vs 1.003 at 128³); $R_{fl}(400)$ is 1.658 (256³) and 1.757 (128³), and the corresponding onset is slightly earlier at 256³ (270 versus 277), rather than evidence of a disagreement between classifiers. Figure labels were checked: every primed ratio in the figures is R_{fl} .

- **Resolution follow-up (interim, added 2026-10-03).** Same-field 128³/256³/512³ initial conditions, a 1024² two-dimensional baseline and an accelerating-frame check are reported in REPORT_resolution.md. The 512³ runs and the 256³ $r = 0.3$ pair are still pending; nothing in this document is revised on their basis yet.

The Vallis-2D report is by the same author as this synthesis; following a second independent review it is corrected here as follows:

- **Deep energy drift.** The source says the Stokes-term residual "vanishes with resolution"; the data show +0.6% (256²) and +0.1% (512²) over 200 T_e , i.e. a decrease between two grids, and this synthesis says so.
- **Ring width.** The source infers a " ≈ 1 -mode" VM93 ring from $Z/E = 2K^2$; that ratio is consistent with a narrow ring but does not determine the width, so $\sigma = 1$ mode is described here as a chosen value with $\sigma = 0.5$ and 2 as sensitivity tests.
- **Deep seed comparison.** Compared at equal times: seed 2 gives $K_{jet}/K = 0.074$ at 100 T_e against ≈ 0.2 for seed 1 at 100 T_e (not against the 0.39 that seed 1 reaches at 200 T_e).
- **Dumbbell overlay.** The dashed curves are a velocity-based Rhines estimate $K^2 = \beta |\cos \theta| / u'_{rms}(t)$; they are not the ε -based wave-turbulence boundary of Vallis Eq. 12.14, whose contour has a different shape. Captions say "Rhines estimate".
- **"Resolution-converged at 256²"** is softened to agreement between 256² and 512²; "reproduced" means a source-matched qualitative comparison, not an exact reproduction.

Paths, commits and data

	A · SURFACE STRESS	B · VALLIS-2D
Code	/u/glwagner/nwt_wc25_surface_stress, branch glw/wc25-surface-stress (HEAD 8c5bc97): experiments/wc25_surface_stress, analysis/wc25_surface_stress, batch/wc25_*.batch	/u/glwagner/nwt_wc25_vallis2d, branch glw/wc25-vallis2d (HEAD f21278d + this document): experiments/wc25_vallis2d, analysis/wc25_vallis2d, batch/vallis_*.batch
Data	/work/hdd/bhcr/glwagner/wc25_surface_stress_2026-10-02/<waves>_r<ratio>_seed<s>_N<N>/	/work/hdd/bhcr/glwagner/wc25_vallis2d_2026-10-02/<waves>_ring_k<k0>_w<sigma>_rb<rb>_seed<s>_N<N>/, betaplane_*/, initial_conditions/
Figures	figures/wc25_surface_stress/{pilot1,sweep1,sweep2,final,final256,final_medium,robust}_*.png, jets_vs_rolls_sections.mp4	figures/wc25_vallis2d/{pilot,final}_*.png, final_metrics.txt
Reports	experiments/wc25_surface_stress/REPORT.md (bundled as REPORT_surface_stress.md)	figures/wc25_vallis2d/REPORT.md (bundled as REPORT_vallis2d.md)
Status / ledger	/u/glwagner/wc25_surface_stress_status.md, wc25_surface_stress_coordination/ledger/case_table.md	/u/glwagner/wc25_vallis2d_status.md, wc25_vallis2d_coordination/{submit_requests,submit_responses}

PNG figures are ignored by the repository's .gitignore and live on disk only (the surface-stress campaign force-added its final set). This document references them by relative path from figures/wc25_results/; the review bundle preserves that layout. Source movies and the pre-existing packet-wake gallery content are served from the published glw/anti-stokes branch and are not duplicated in the bundle.

References

- Wagner & Constantinou (2025), *Phenomenology of decaying turbulence beneath surface waves*, JFM 1020, A51. doi:10.1017/jfm.2025.10649.
- Vallis & Maltrud (1993), *Generation of Mean Flows and Jets on a Beta Plane and over Topography*, JPO 23, 1346–1362. Author PDF.
- Vallis (2017), *Atmospheric and Oceanic Fluid Dynamics*, second edition, Cambridge University Press, Figs 11.8, 12.3 and 12.4. Related author teaching code.